

PIZZA SALES ANALYTICS REPORT

January 2015 – December 2015

PROJECT DETAILS

Tools: Microsoft Power BI | Microsoft SQL Server
Dataset: Pizza Sales — Jan 2015 to Dec 2015
Type: Business Intelligence & Data Analytics

\$817.86K

Total Revenue

\$38.31

Avg Order Value

49,574

Pizzas Sold

21,350

Total Orders

2.32

Avg Pizzas/Order

1. EXECUTIVE SUMMARY

This report presents a comprehensive analysis of pizza sales performance for the full calendar year January 2015 through December 2015. The data was processed and visualized using Microsoft Power BI and SQL, producing interactive dashboards that covers key performance indicators, order trend analysis, category and size breakdowns, and product-level best and worst sellers.

The analysis draws on transaction-level order data encompassing over 21,000 orders and nearly 50,000 individual pizzas sold. The insights derived are intended to support strategic decisions in menu management, pricing, staffing, and marketing investment.

1.1 Dashboard Overview

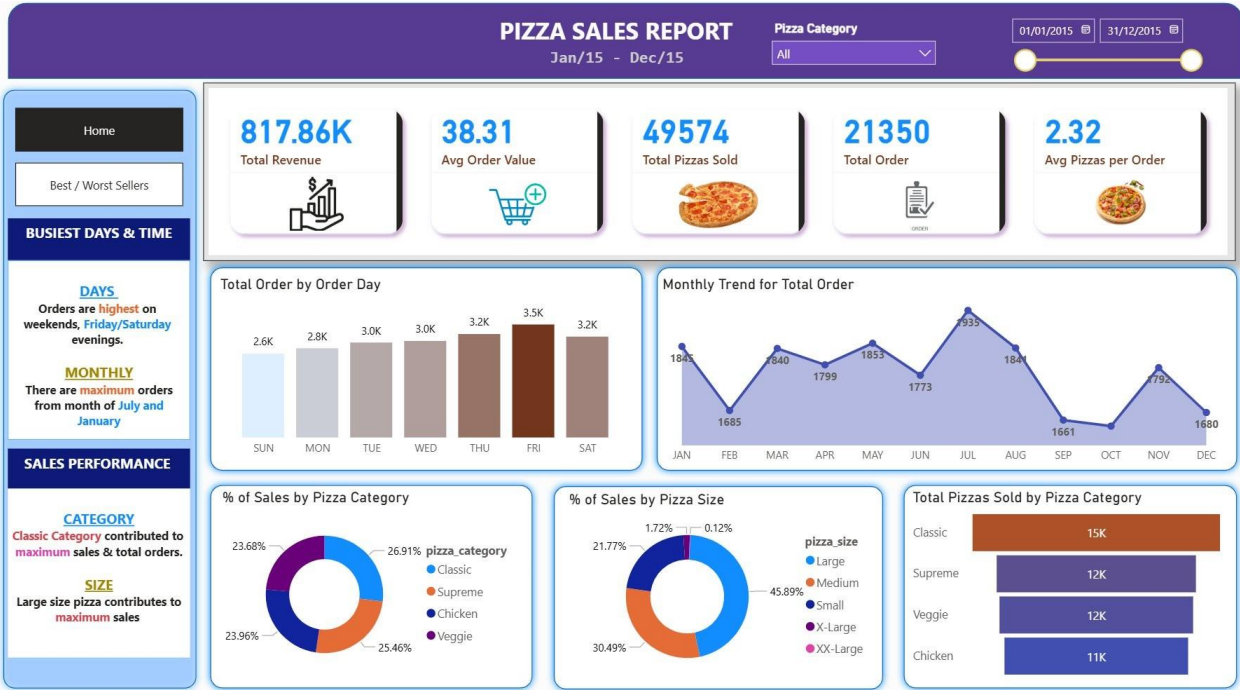


Figure 1 — Power BI Home Dashboard: KPIs, Order Trends, and Category Breakdown (Jan-Dec 2015)



2. KEY PERFORMANCE INDICATORS

The five core KPIs defined in the project brief have been computed and are summarised below.

\$817.86K	\$38.31	49,574	21,350	2.32
Total Revenue	Avg Order Value	Total Pizzas Sold	Total Orders	Avg Pizzas/Order

Total Revenue: \$817,860

The business generated \$817,860 in total revenue across all orders in 2015. This is a strong commercial result and forms the foundation for evaluating per-order and per-product profitability.

Average Order Value: \$38.31

Customers spent an average of \$38.31 per order. This metric is important for pricing strategy and promotional planning. An Avg Order Value of \$38 suggests customers are typically ordering multiple items or premium products, consistent with the average of 2.32 pizzas per order.

Total Pizzas Sold: 49,574

49,574 pizzas were sold during the year, indicating high output. This figure supports capacity planning for ingredients, production, and delivery logistics.

Total Orders: 21,350

The business fulfilled 21,350 orders across the year, averaging roughly 58 orders per day. Order volume shows meaningful variation across days and months, detailed in Section 3.

Average Pizzas Per Order: 2.32

Customers ordered an average of 2.32 pizzas per transaction. This signals a strong tendency toward shared meals or group orders, which has implications for bundle promotions.



3. ORDER TREND ANALYSIS

3.1 Daily Trend for Total Orders

The bar chart on the dashboard (Figure 1, lower-left panel) displays total orders aggregated by day of the week. The data reveals a clear and consistent weekly pattern:

- Friday is the busiest day with approximately 3,500 orders, the weekly peak.
- Saturday follows closely with around 3,200 orders, confirming strong weekend demand.
- Sunday records the lowest volume at approximately 2,600 orders, suggesting a slower start to the week.
- Mid-week (Tuesday through Thursday) shows stable moderate volumes in the range of 2,800–3,200 orders.

The Friday–Saturday surge is consistent with typical restaurant and takeaway industry patterns, where customers are more inclined to dine out or order in at the end of the working week. Operationally, this indicates the need for higher staffing levels and ingredient pre-preparation heading into the weekend.

3.2 Monthly Trend for Total Orders

The line chart (Figure 1, lower-right panel) tracks total monthly order volumes throughout the year. Key observations include:

- July is the highest-performing month at 1,935 orders, driven likely by summer demand and social gatherings.
- January (1,845 orders) also performs strongly, possibly fueled by New Year promotions and post-holiday ordering habits.
- February records the year's lowest monthly volume at 1,685 orders, a dip consistent with the short month and post-January quieter period.
- September (1,661 orders) represents the second-lowest point, marking a notable late-summer lull before the year-end recovery.
- Order volumes recover through October and November before dipping again in December (1,680), which may reflect customers opting for home-cooked holiday meals.

The seasonal rhythm observed suggests the business could benefit from targeted mid-winter and mid-autumn promotions to lift volumes during the troughs in February and September.

4. SALES BY CATEGORY AND SIZE

4.1 Percentage of Sales by Pizza Category

The donut chart in the dashboard shows how total sales are distributed across the four pizza categories: Classic, Supreme, Chicken, and Veggie.

Pizza Category	% of Sales	Observation
Classic	26.91%	Highest — dominant category
Supreme	25.46%	Second — close competitor
Chicken	23.96%	Third — strong performance
Veggie	23.68%	Lowest — nearly equal share

The near-equal distribution across categories (ranging only 3.2 percentage points from top to bottom) suggests a broadly balanced menu with no single category dominating. Classic pizza holds a slight edge, but the business does not rely on any one category, which reduces risk from changing consumer preferences.

The funnel chart (Figure 1, bottom-right) reinforces this picture in absolute quantity: Classic leads at 15K units, followed by Supreme and Veggie at 12K each, and Chicken at 11K.

4.2 Percentage of Sales by Pizza Size

The second donut chart reveals a strong customer preference skew by size:

Pizza Size	% of Sales	Observation
Large	45.89%	Most popular — nearly half of all sales
Medium	30.49%	Second — substantial volume
Small	21.77%	Third — moderate share
X-Large	1.72%	Niche — very limited demand
XX-Large	0.12%	Minimal — negligible share

Large pizzas account for nearly half of all sales, making it by far the most popular size. Combined with Medium, the two sizes cover over 76% of all orders. X-Large and XX-Large together account for less than 2% of sales, suggesting these sizes may not warrant the same level of inventory investment as the core Large and Medium offerings.



5. BEST AND WORST SELLERS

The second dashboard page (Figure 2) provides a detailed breakdown of pizza performance at the individual product level, ranked across three dimensions: Revenue, Total Quantity Sold, and Total Orders.

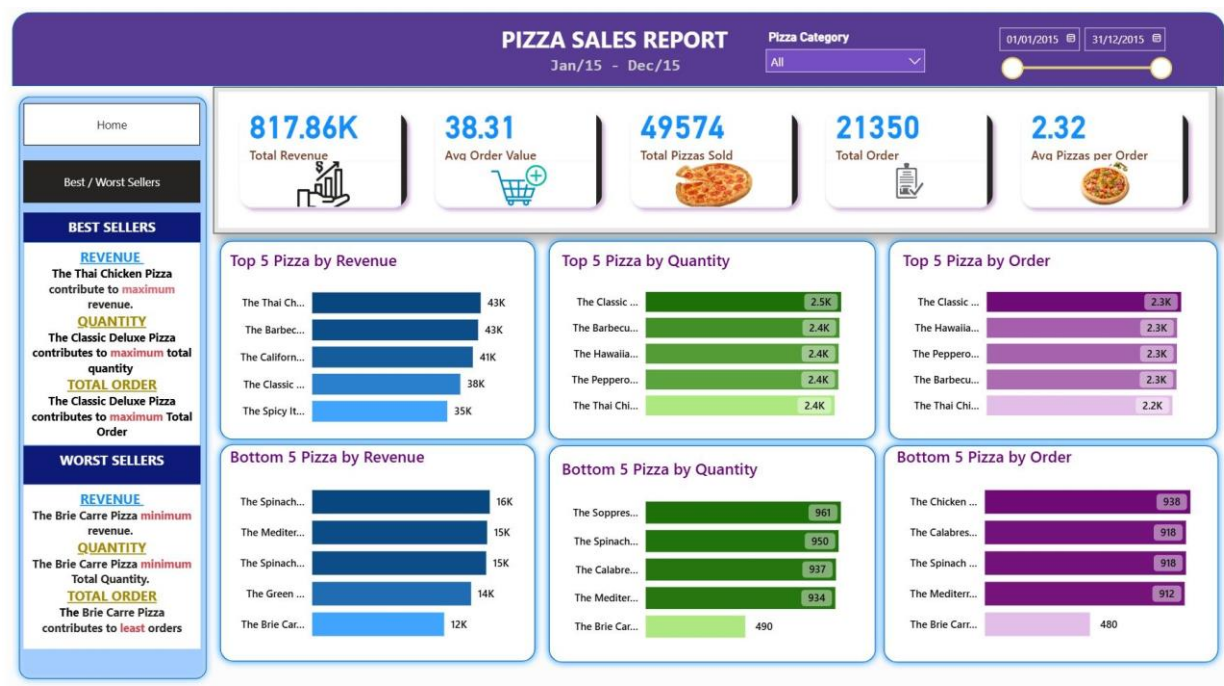


Figure 2: Power BI Best & Worst Sellers Dashboard: Top 5 and Bottom 5 Pizzas by Revenue, Quantity, and Orders



5.1 Top 5 Best Sellers

By Revenue

The Thai Chicken Pizza and The Barbecue Chicken Pizza are joint top earners, each generating approximately \$43K in revenue. The California Chicken Pizza follows at \$41K, with The Classic Deluxe Pizza at \$38K and The Spicy Italian Pizza rounding out the top five at \$35K.

The dominance of chicken-based pizzas in the revenue ranking suggests these products command higher price points or larger portion sizes, making them highly valuable to the top line despite not necessarily being the most frequently ordered.

By Quantity Sold

The Classic Deluxe Pizza leads in volume at approximately 2,500 units, closely followed by The Barbecue Chicken, The Hawaiian, The Pepperoni, and The Thai Chicken, all clustered around 2,400 units. The tight grouping in the top five indicates that these pizzas enjoy consistently broad customer appeal across the full year.

By Total Orders

The Classic Deluxe and The Hawaiian Pizza share the top spot for most frequently ordered, each appearing in approximately 2,300 orders. The Pepperoni, Barbecue Chicken, and Thai Chicken complete the top five, all at around 2,200–2,300 orders. The Classic Deluxe Pizza's consistent appearance across all three rankings confirms it as the most universally popular product in the range.

5.2 Bottom 5 Worst Sellers

By Revenue

The Brie Carre Pizza is the lowest revenue generator at \$12K, less than a third of the top performers. The Green Garden Pizza (\$14K), The Spinach Supreme (\$15K), and The Mediterranean Pizza (\$15K) also underperform significantly on revenue.

By Quantity and Orders

The Brie Carre Pizza is also the worst performer by both quantity (490 units) and orders (480), making it the single most underperforming product in the portfolio across all three dimensions. The Soppressata, Spinach, Calabrese, and Mediterranean pizzas also consistently feature in the bottom five, suggesting persistent low demand rather than seasonal weakness.

The sharp contrast between top and bottom performers, with the best-selling pizza outselling the worst by more than five times in unit volume, highlights a long-tail product distribution common in food service. The business should evaluate whether the bottom performers contribute sufficient margin to justify their continued production complexity and ingredient overhead.



6. KEY INSIGHTS AND RECOMMENDATIONS

1. Capitalise on the Friday–Saturday Peak

With Friday generating the highest single-day order volume (3,500 orders), targeted weekend promotions, bundle deals, and pre-orders could further amplify revenue on peak days. Ensuring adequate staffing and ingredient stock for Thursday through Saturday is operationally critical.

2. Address the February and September Troughs

Order volumes dip meaningfully in February and September. Loyalty incentives, limited-edition seasonal menu items, or discount promotions during these months could help sustain revenue through the quieter periods.

3. Double Down on Large Pizzas

Large pizzas represent 45.89% of all sales. Ensuring competitive pricing, high availability, and strong promotion of Large-size options will continue to drive the majority of revenue. The X-Large and XX-Large tiers may benefit from review to determine whether they are worth maintaining.

4. Promote the Classic Category Strategically

While Classic holds the largest sales share at 26.91%, the four categories are closely matched. Category-specific promotions (e.g., a 'Veggie Week' campaign) could be used to grow the lower-share categories or defend the Classic lead.

5. Leverage the Thai Chicken and Classic Deluxe as Hero Products

The Thai Chicken Pizza leads on revenue while the Classic Deluxe leads on quantity and orders. Both should be prominently featured in marketing materials, digital menus, and loyalty reward programs.

6. Review or Retire the Brie Carre Pizza

The Brie Carre Pizza consistently underperforms across revenue, quantity, and orders. A full profitability review, including ingredient cost, preparation time, and customer feedback, should be conducted to determine whether this product should be reformulated, repositioned, or removed.

7. Explore Upselling to Increase AOV

With an Average Order Value of \$38.31 and an average basket of 2.32 pizzas, there is room to grow AOV through strategic upselling of sides, drinks, or desserts. Bundle promotions pairing top-selling pizzas with complementary items could increase transaction value without requiring new customer acquisition.

7. MICROSOFT SQL SERVER — QUERY DOCUMENTATION

Prior to building the Power BI dashboard, all KPIs and chart metrics were validated using Microsoft SQL Server. Structured queries were written against the `pizza_sales` table to extract, aggregate, and verify each metric. This section documents every SQL query used, accompanied by screenshots of the query output directly from SQL Server Management Studio (SSMS).



7.1 KPI Queries

Total Revenue

```
SELECT SUM(total_price) AS Total_Revenue
FROM pizza_sales
```

Returns the sum of the `total_price` column across all rows in the `pizza_sales` table, giving the overall revenue for the period.

Average Order Value

```
SELECT SUM(total_price) / COUNT(DISTINCT order_id) AS Average_Order_Value
FROM pizza_sales
```

Uses `DISTINCT` on `order_id` to count unique orders only, avoiding inflation from multiple line items per order (e.g. Order 2,2,2,2 counted as one, not four).

Total Pizzas Sold

```
SELECT SUM(quantity) AS Total_Pizza_Sold
FROM pizza_sales
```

Sums the `quantity` column to get the total number of individual pizzas sold.

Total Orders

```
SELECT COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
```

Counts unique order IDs to return the total number of distinct customer orders placed.

Average Pizzas Per Order

```
SELECT CAST(SUM(quantity) AS DECIMAL(10,2)) / COUNT(DISTINCT order_id)
      AS Average_Pizzas_Per_Order
FROM pizza_sales
```

Casts the `quantity` sum to `DECIMAL` before dividing to ensure a precise decimal result rather than integer division.

7.2 Chart Queries

Daily Trend for Total Orders

```
SELECT DATENAME(DW, order_date) AS Order_Day,  
       COUNT(DISTINCT order_id) AS Total_Orders  
FROM pizza_sales  
GROUP BY DATENAME(DW, order_date)
```

DATENAME(DW, ...) extracts the full weekday name from the order date. Results are grouped by day name to produce order counts for each day of the week.

Monthly Trend for Total Orders

```
SELECT DATENAME(MONTH, order_date) AS Month_Name,  
       COUNT(DISTINCT order_id) AS Total_Orders  
FROM pizza_sales  
GROUP BY DATENAME(MONTH, order_date)  
ORDER BY Total_Orders DESC
```

DATENAME(MONTH, ...) extracts the full month name. The ORDER BY DESC clause ranks months from highest to lowest order volume, making it easy to identify peak and trough months.

Top 5 Best Sellers — By Revenue

```
SELECT TOP 5 pizza_name,  
            SUM(total_price) AS Total_Revenue  
FROM pizza_sales  
GROUP BY pizza_name  
ORDER BY Total_Revenue DESC
```

Ranks pizzas by total revenue and returns only the top five using the TOP 5 clause.

Top 5 Best Sellers — By Quantity

```
SELECT TOP 5 pizza_name,  
            SUM(quantity) AS Total_Quantity  
FROM pizza_sales  
GROUP BY pizza_name  
ORDER BY Total_Quantity DESC
```

Ranks by total quantity sold to identify the five most frequently ordered pizzas by volume.

Top 5 Best Sellers — By Total Orders

```
SELECT TOP 5 pizza_name,  
            COUNT(DISTINCT order_id) AS Total_Orders  
FROM pizza_sales  
GROUP BY pizza_name  
ORDER BY Total_Orders DESC
```

Counts distinct orders per pizza name to identify which products appear most frequently across unique customer transactions.

7.3 SQL Server Output Screenshots

The following screenshots show the SQL queries as executed in Microsoft SQL Server Management Studio (SSMS), along with their result sets. These outputs were used to cross-validate all KPI values displayed in the Power BI dashboard.

PIZZA SALES SQL QUERIES

A. KPI's

1. Total Revenue

```
SELECT SUM(total_price) AS Toatal_Revenue from pizza_sales
```

Results		Messages	
		Toatal_Revenue	
1		817860.05083847	

2. Average Order Values

```
SELECT SUM(total_price) / COUNT (DISTINCT order_id) AS Average_Order from pizza_sales
```

(Order_id must be distinct to get the actual order, it is done to avoid repeating counting of the same order eg. Order 2,2,2,2,2 as 5 different orders)

Results		Messages	
		Average_Order	
1		38.3072623343546	

3. Total Pizza Sold

```
SELECT SUM(quantity) AS Total_Pizza_Sold from pizza_sales
```

Results		Messages	
		Total_Pizza_Sold	
1		49574	

4. Total Orders

```
SELECT COUNT(DISTINCT order_id ) AS Total_orders from pizza_sales
```

Results		Messages	
		Total_orders	
1		21350	

5. Average Pizzas per Order (Total Pizzas / no. of Orders)



Figure 3 — SQL Server: KPI Queries (Total Revenue, AOV, Total Pizzas Sold)

```
SELECT SUM(quantity) / COUNT(DISTINCT order_id) AS Average_pizzas_per_order
```

Results Messages	
Average_pizzas_per_order	
1	2

```
SELECT CAST(SUM(quantity) AS DECIMAL(10,2)) / COUNT(DISTINCT order_id) AS  
Average_pizzas_per_order from pizza_sales
```

Results Messages	
Average_pizzas_per_order	
1	2.3219672131147

B. CHARTS REQUIREMENTS

1. Daily Trend For Total Orders

```
SELECT DATENAME(DW, order_date) as order_day, COUNT(DISTINCT order_id) AS  
Total_Orders FROM pizza_sales  
GROUP BY DATENAME(DW, order_date)
```

	order_day	Total_Orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

2. Monthly Trend for Total Orders

```
SELECT DATENAME(MONTH, order_date) AS Month_Name, COUNT(DISTINCT order_id) AS Total_Orders  
FROM pizza_sales  
GROUP BY DATENAME(MONTH, order_date)
```



Figure 4 — SQL Server: KPI Queries continued (Total Orders, Avg Pizzas per Order)

Results Messages		
	Month_Name	Total_Orders
1	February	1685
2	June	1773
3	August	1841
4	April	1799
5	May	1853
6	December	1680
7	January	1845
8	September	1661
9	October	1646
10	July	1935
11	November	1792
12	March	1840

```

SELECT DATENAME(MONTH, order_date) AS Month_Name, COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY DATENAME(MONTH, order_date)
ORDER BY Total_Orders DESC

```

	Month_Name	Total_Orders
1	July	1935
2	May	1853
3	January	1845
4	August	1841
5	March	1840
6	April	1799
7	November	1792
8	June	1773
9	February	1685
10	December	1680
11	September	1661
12	October	1646

(TO WRITE IN DESCENDING ORDER)

Figure 5 — SQL Server: Chart Queries (Daily Trend, Monthly Trend)

3. Percentage of Sales by Pizza Category

```

2 SELECT
3     pizza_category,
4     SUM(total_price) * 100.0 / (SELECT SUM(total_price) FROM pizza_sales) AS percentage
5 FROM pizza_sales
6 GROUP BY pizza_category;

```

100 % No issues found

Results Messages

	pizza_category	percentage
1	Classic	26.9059602306976
2	Chicken	23.9551375322885
3	Veggie	23.6825910258677
4	Supreme	25.4563112111462

4. Percentage of Sales by Pizza Size

```

1 SELECT
2     pizza_size,
3     SUM(total_price) * 100.0 / (SELECT SUM(total_price) FROM pizza_sales) AS percentage
4 FROM pizza_sales AS Total_Sales
5 GROUP BY pizza_size;

```

100 % No issues found

Results Messages

	pizza_size	percentage
1	L	45.8903330244889
2	XXL	0.123077294254725
3	M	30.482044420569
4	XL	1.72107684995364
5	S	21.7734684107037

5. Top 5 Best Sellers by Revenue, Total Quantity, and Total Orders

```

SELECT TOP 5 pizza_name, SUM(total_price) AS Total_Revenue From pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue DESC

```

Results Messages

	pizza_name	Total_Revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

```

SELECT TOP 5 pizza_name, SUM(quantity) AS Total_Quantity From pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity DESC

```

Figure 6 — SQL Server: Chart Queries (Category %, Size %, Top 5 Best Sellers)

Results Messages		
	pizza_name	Total_Quantity
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

SELECT TOP 5 pizza_name, COUNT(DISTINCT order_id) AS Total_Orders From pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders DESC

Results Messages		
	pizza_name	Total_Orders
1	The Classic Deluxe Pizza	2329
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Pizza	2273
5	The Thai Chicken Pizza	2225



Figure 7 — SQL Server: Bottom 5 Worst Sellers Queries

8. CONCLUSION

The 2015 pizza sales data reveals a business with solid fundamentals: strong overall revenue of \$817,860, consistent order volumes averaging and over 58 transactions per day. This project employed a two-tool analytical approach; Microsoft SQL Server for data extraction and metric validation, and Microsoft Power BI for interactive visualisation and dashboard delivery, ensuring both accuracy and accessibility of the insights produced.

The SQL queries documented in Section 7 confirm the integrity of every KPI and chart metric visible in the Power BI dashboard. The analysis reveals several actionable patterns: the Friday revenue peak, the July seasonal high, the clear dominance of Large-size pizzas, and the consistent underperformance of specific products. Acting on these findings through targeted promotions, menu rationalisation, and capacity optimisation represents a meaningful opportunity to improve both revenue and operational efficiency.

Further analysis recommended for future iterations includes year-on-year comparison once multi-year data is available, customer segmentation by order frequency and spend, geographic heat mapping if location data is present, and a full product profitability model incorporating cost of goods sold.

End of Report — Pizza Sales Analytics | Jan–Dec 2015 | Microsoft SQL Server & Power BI